

Submission PHYS1-PS1-SUB08

Question PHYS1-PS1-Q01

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visible work:

$$v = v_{\text{initial}} + at = 3 + 2(5) = 13 \text{ m/s} \dots ?$$

last line is hard to read; maybe $v = 13 \text{ m/s}$; displacement = 40 m

Question PHYS1-PS1-Q02

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visible work:

$$N = mg = 29.4 \text{ N}; \text{ friction} = \mu N = 5.9 \text{ N} \dots ?$$

last line is hard to read; maybe $a = 4.04 \text{ m/s}^2$; friction opposes motion

Question PHYS1-PS1-Q03

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visible work:

$$mgh = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{(2gh)} = \sqrt{(2 \cdot 9.8 \cdot 7)} \dots ?$$

last line is hard to read; maybe $v = 11.71 \text{ m/s}$

Question PHYS1-PS1-Q04

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visible work:

$$p_{\text{total, before}} = p_{\text{total, after}} \dots ?$$

last line is hard to read; maybe total momentum before equals after when external impulse is negligible

Question PHYS1-PS1-Q05

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visible work:

$$\text{Impulse is area under } F(t): J = F\Delta t = 6(0.5) = 3 \text{ N}\cdot\text{s} \dots ?$$

last line is hard to read; maybe impulse = 3 N·s; momentum changes by 3 kg·m/s

Question PHYS1-PS1-Q06

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visible work:

$$v = v_{\text{initial}} + at = 4 + 4(3) = 16 \text{ m/s} \dots ?$$

last line is hard to read; maybe $v = 16 \text{ m/s}$; displacement = 30 m